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NutriMet

Context-Aware Daily Calorie & Energy Expenditure Estimator

Individual Capstone Project Proposal

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Status: project proposal, system design and an initial UI prototype. The full application is planned for future development.

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Abstract

Most calorie calculators estimate daily energy needs with a single fixed activity multiplier. This ignores the real context of a person's day, such as how long they slept, how hot it was, how stressed they felt and which activities they actually did. NutriMet is a simple, context-aware estimator that combines basal metabolic rate (BMR), activity energy based on MET values, a context adjustment factor and the thermic effect of food (TEF) to give a more realistic daily energy expenditure figure. The project follows an Incremental Model and is scheduled over six weeks. The prototype uses Python with Flask, SQLite and a simple HTML/CSS front end. This document contains the requirements, design diagrams, database design, testing strategy and an initial UI prototype plan.

1 Introduction and Background

Students and young adults often track calories using generic online calculators. These tools multiply BMR by a fixed factor (sedentary, moderate, active) and give one number for every day. In reality, energy use changes from day to day depending on movement, sleep, weather and stress. Studies on resting energy expenditure and the Compendium of Physical Activities provide well-established formulas and MET values that can be combined into a better estimate.

NutriMet proposes a small application that takes these inputs together and explains the result clearly, so that the user can understand where the calories go each day.

2 Problem Statement

Existing calorie calculators have several weaknesses:

- they use one fixed activity multiplier for all days;
- they ignore context such as sleep, temperature and stress;
- they rarely show how the final number was calculated;
- they do not keep a daily history that lets users compare their days.

As a result, users get estimates that are too general to be useful for daily planning.

3 Aims and Objectives

Aim: To design, and in later weeks implement, a simple web application that estimates daily calorie and energy expenditure by considering the user's body profile, daily activities and context.

Objectives:

1. Analyse and document the requirements of the User and Administrator roles.
2. Produce the system design: DFD, use case, activity, ERD, class and object diagrams.
3. Design a small relational database for users, daily logs, activities, MET values and estimates.
4. Implement the BMR, activity energy, context factor and TEF calculations.
5. Implement profile, daily logging and history features with Flask and SQLite.
6. Provide an administrator interface for managing MET values and users.
7. Test the main workflow and deliver the project on GitHub.

4 Project Scope

Included	Excluded
User registration and login with two roles	Medical diagnosis or treatment advice
Body profile management (age, sex, height, weight)	Wearable device or sensor integration
Daily activity logging with durations	Automatic food image recognition
Context input: temperature, sleep, stress	Meal planning and diet recommendation
BMR, activity, context factor and TEF calculation	Mobile native applications
Estimate history and simple charts	Social features and sharing
Admin management of MET values and users	Cloud deployment and microservices

5 Stakeholders and User Roles

- **User:** registers, keeps a profile, logs daily activities and context, and views estimates and history.
- **Administrator:** manages the activity/MET list and user accounts.
- **Supervisor/Evaluator:** reviews the proposal, design and prototype.
- **Developer:** the student who designs, builds and tests the system.

6 Functional Requirements

ID	Requirement
FR1	The system shall allow a user to register, log in and log out.
FR2	The user shall be able to create and update a profile (age, sex, height, weight).
FR3	The user shall be able to select activities and enter durations for a day.
FR4	The user shall be able to enter context: temperature, sleep hours and stress level.
FR5	The system shall calculate BMR using the Mifflin-St Jeor equation.
FR6	The system shall calculate activity energy using $(\text{MET}-1)\times weight\times hours$.
FR7	The system shall apply a context factor and TEF to produce total daily energy expenditure.
FR8	The system shall display a breakdown of BMR, activity, context and TEF.
FR9	The system shall save every estimate and show the user's history.
FR10	The administrator shall add, edit and remove activities and MET values.
FR11	The administrator shall view and manage user accounts.

7 Non-Functional Requirements

- **Usability:** simple forms and a clear result page that can be used without training.
- **Performance:** an estimate shall be produced in under two seconds.
- **Security:** passwords are stored as hashes; admin pages are role-protected.

- **Reliability:** input validation prevents impossible values (for example negative weight).
- **Maintainability:** calculation logic is separated from the user interface.
- **Portability:** runs in any modern browser.

8 Feasibility Study

Technical: Python, Flask and SQLite are free, well documented and suitable for a student project. **Economic:** no paid tools or services are needed. **Operational:** the application is easy to use and can run on a normal laptop. **Schedule:** the scope is limited to a core workflow that can be completed in six weeks.

9 Software Development Methodology

The project follows the **Incremental Model**. Each increment delivers a working part of the system: first the calculation engine, then the database and profile, then logging and history, and finally the admin module and testing. This allows early testing and keeps the scope manageable for one developer.

10 Proposed System Architecture and Technology Stack

NutriMet uses a three-layer architecture: a presentation layer (HTML, CSS, JavaScript), an application layer (Python/Flask with a separate estimator module) and a data layer (SQLite).

Layer	Technology
Front end	HTML5, CSS3, JavaScript (Chart.js)
Back end	Python 3, Flask
Database	SQLite
Testing	pytest
Version control	Git and GitHub

The core estimate is

$$\text{TDEE} = (\text{BMR} + E_{act}) \times C + \text{TEF}, \qquad \text{TEF} = 0.10 \times (\text{BMR} + E_{act}) \times C,$$

where $E_{act} = \sum(\text{MET}_i - 1) \times w \times h_i$ and C is the product of the temperature, sleep and stress factors.

11 System Design

11.1 Data Flow Diagram (DFD)

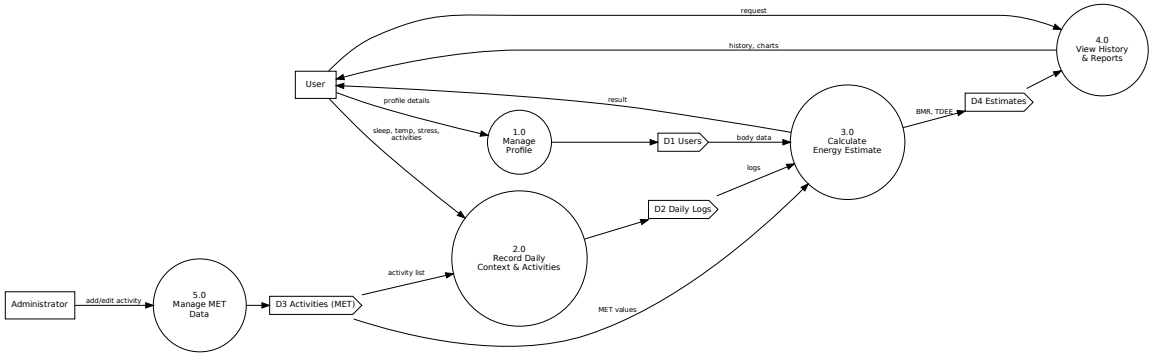


Figure 1: Data flow diagram of NutriMet.

11.2 Use Case Diagram

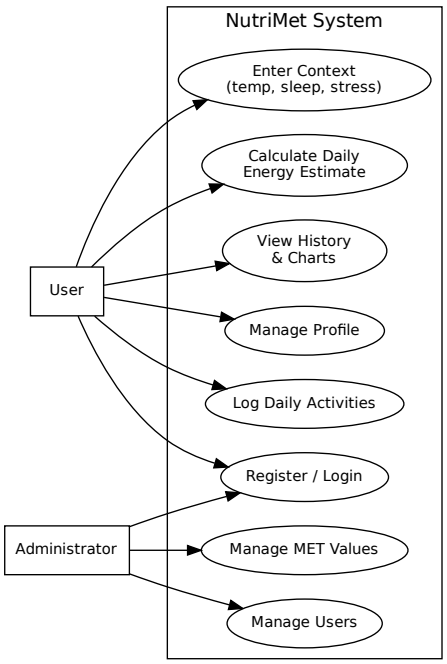


Figure 2: Use case diagram for User and Administrator.

11.3 Activity Diagram

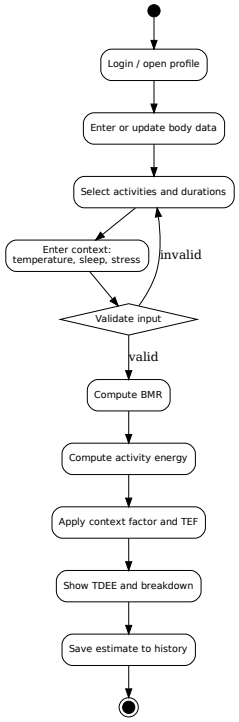


Figure 3: Activity diagram for generating a daily estimate.

11.4 Entity Relationship Diagram (ERD)

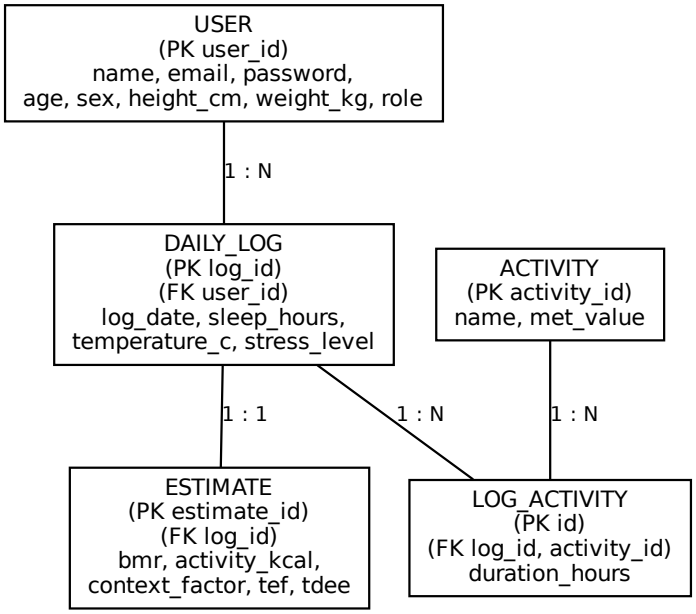


Figure 4: Entity relationship diagram.

11.5 Class Diagram

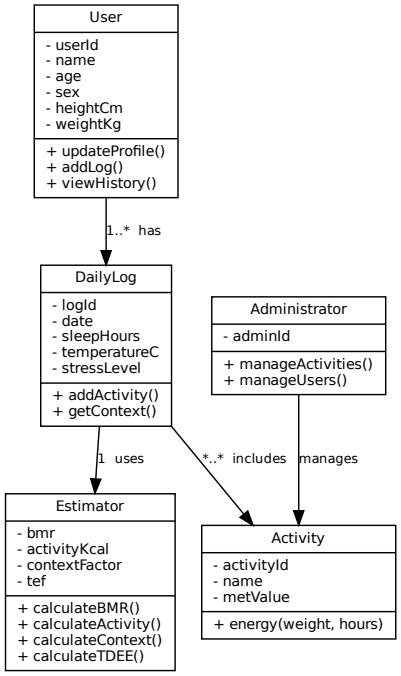


Figure 5: Class diagram.

11.6 Object Diagram

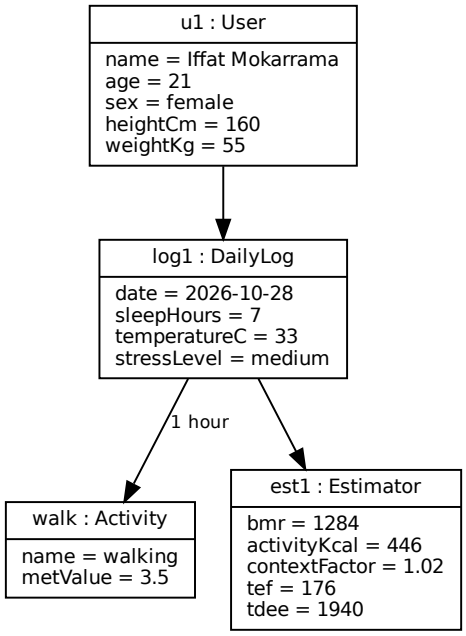


Figure 6: Object diagram for a sample estimate.

12 Basic Database Design

Table	Main fields
users	user_id (PK), name, email, password_hash, age, sex, height_cm, weight_kg, role
activities	activity_id (PK), name, met_value
daily_logs	log_id (PK), user_id (FK), log_date, sleep_hours, temperature_c, stress_level
log_activities	id (PK), log_id (FK), activity_id (FK), duration_hours
estimates	estimate_id (PK), log_id (FK), bmr, activity_kcal, context_factor, tef, tdee

13 UI Prototype Overview

The initial prototype contains the following screens:

- **Login/Register:** simple form with role-based redirect.
- **Profile:** age, sex, height and weight entry.
- **Daily Log:** activity selection with durations and context inputs.
- **Result:** total daily energy with a breakdown of BMR, activity, context and TEF.
- **History:** list and chart of previous estimates.
- **Admin Panel:** manage MET values and users.

14 Six-Week Development Schedule

Week	Focus	Tasks
1	Planning	Requirements, proposal, GitHub repository setup
2	Design	DFD, use case, ERD, class diagrams, UI sketches
3	Core engine	BMR, MET activity energy, context factor, TEF, unit tests
4	Database and UI	SQLite schema, Flask routes, profile and daily log pages
5	History and admin	Result page, history charts, admin MET management
6	Testing and final	System testing, bug fixing, documentation, final submission

15 Milestones and Deliverables

Milestone	Deliverable
End of Week 2	Proposal and design diagrams
End of Week 3	Working calculation engine with tests
End of Week 5	Functional web prototype
End of Week 6	Final system, report and GitHub repository

16 Basic Testing Strategy

- **Unit testing:** BMR, activity energy, context factor and TDEE functions are tested with pytest.
- **Integration testing:** form input to database to result page.
- **Validation testing:** invalid or extreme inputs are rejected with clear messages.
- **User acceptance:** classmates try the prototype and give feedback.

17 Risk Analysis and Mitigation

Risk	Mitigation
Estimates are mistaken for medical advice	Show a clear disclaimer on the result page
Context factors are approximate	Keep them configurable and document the assumptions
Limited time for one developer	Fixed scope and incremental delivery
Data loss	Regular Git commits and database backups

18 Expected Outcomes

At the end of the project NutriMet will provide a working, easy-to-use estimator that gives a more realistic daily calorie and energy expenditure figure than a fixed multiplier, with a clear breakdown, stored history and complete design documentation.

19 Conclusion

NutriMet addresses a practical problem with a small, realistic scope. By combining BMR, MET-based activity energy, daily context and TEF, it produces an explainable estimate. The proposal, system design and prototype plan prepare the ground for full implementation in the coming weeks.

References

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